

Active TAWERS 4 WG4

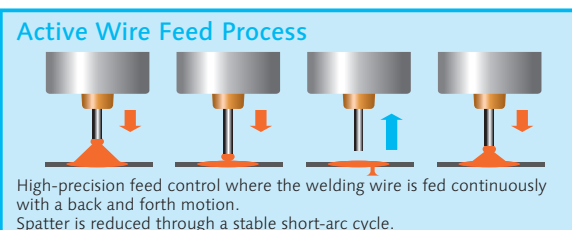
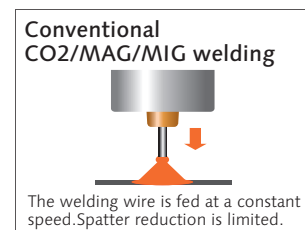
The welding power source integrated robot has evolved into a new range, achieving high-speed and ultra-low-spatter welding

Active Wire Feed Process 4 (AWP4) AWP4(Active Wire Feed Process 4)

Wider current range and precise wire feed

- Contribute to productivity improvement with high-speed welding and ultra-low spatter
- Achieve 100% duty cycle at 310 A

(When using CO₂ gas, 1.2 mm mild steel solid wire, and an air-cooling unit)



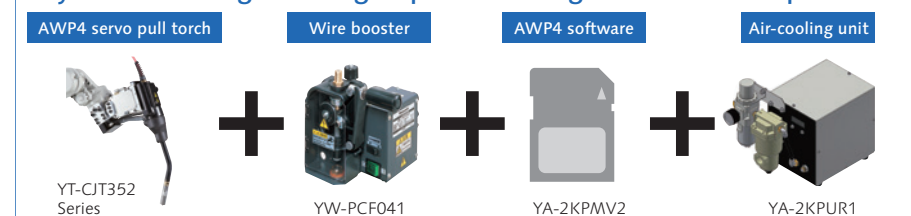
WG4

TS	TM	TL	LA
800	1100	1800	1800
950	1400	2000	
	1600		
	1800		
	2000		

* TS:Through-arm/External
* TM:Separate/Through-arm
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System enabling both high-speed welding and ultra-low spatter



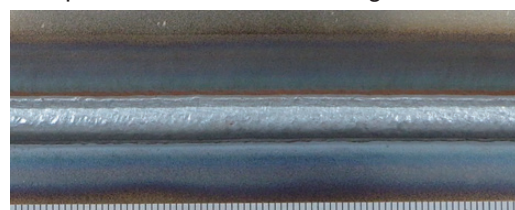
Please contact us for details.

High-speed welding

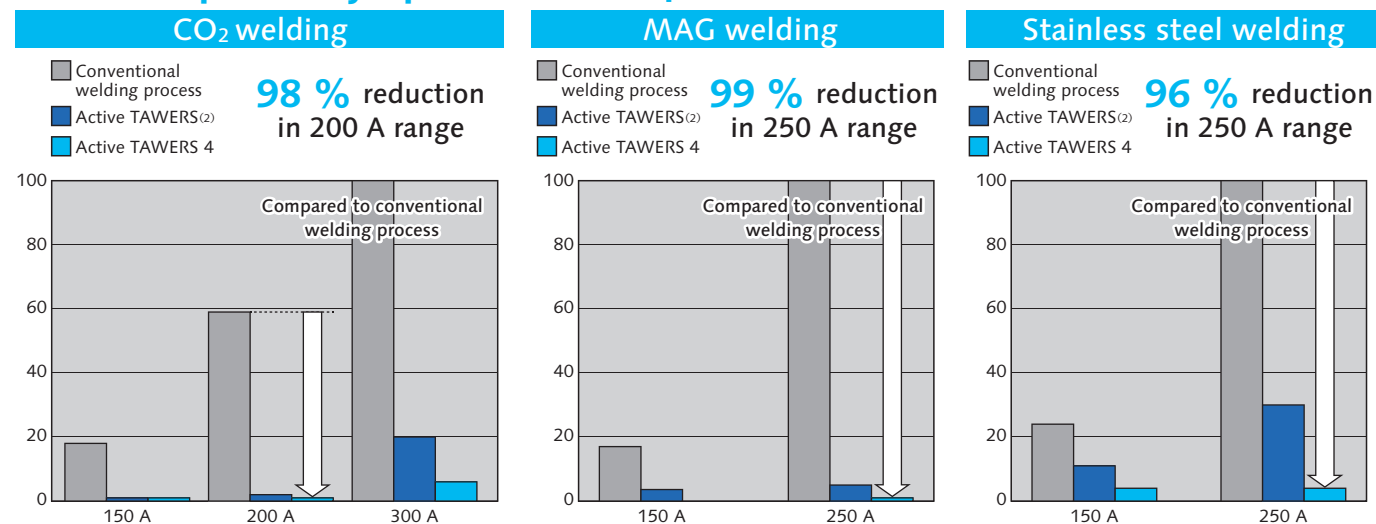
- Productivity improved at speeds of 100 cm/min or higher
- Smooth and beautiful bead appearance

Welding conditions: Joint: Lap Gas: CO₂
Welding current: 320 A
Welding speed: 110 cm/min
Plate thickness: 3.2 mm

Example of mild steel SPCC welding



Reduces spatter by up to 99% (compared to conventional models)



Note: Precautions for using Active TAWERS 4 servo pull torch
1. Use a coated pail-pack wire. (Panasonic wire recommended.)
2. Adjust the wire cast diameter to 1000 to 1200 mm.

Active TAWERS 4 WG4

Burn-through prevention, higher gap tolerance, and better bead appearance Applicable to wider ranges

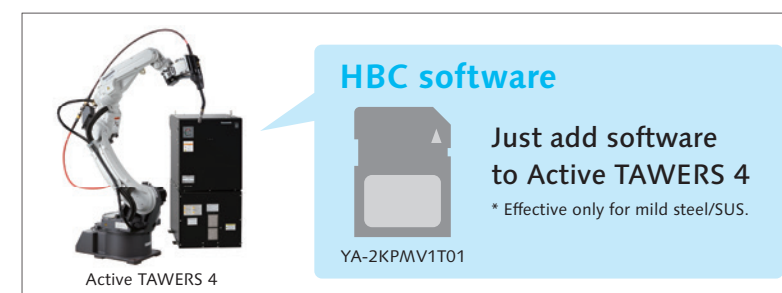
Active Wire Feed Process (Optional for thin-plate and gap welding)

HBC(Heat Balance Control) process supports welding of high-tensile steel plates that are becoming thinner

WG4

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800	1100	1800	1800
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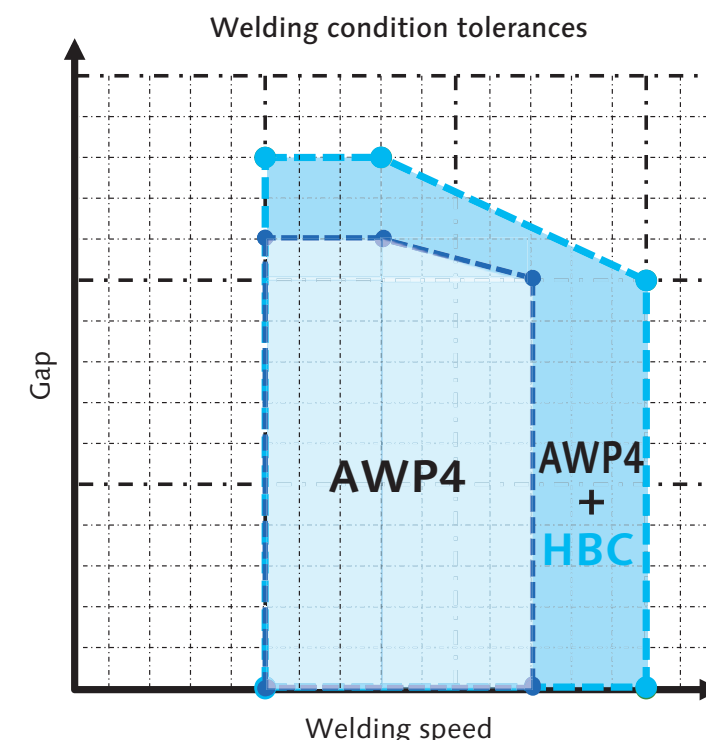
Suppresses burn through in thin plate welding

- Low heat input control significantly increases condition tolerances (welding speeds and gaps)
- Capable of welding thin high-tensile steel plates that are prone to burn-through

Example of high-tensile steel (980 MPa)



Welding conditions: Joint: Lap Gas: MAG
Welding current: 150 A
Welding speed: 100 cm/min
Plate thickness: 0.8 mm Gap: 1 mm



Conventional S-AWP basic functions are included in the AWP4 software (YA-2KPMV2).

Note: Precautions for using AWP4
1. Use a coated pail-pack wire. (Panasonic wire recommended.)
2. Adjust the wire cast diameter to 1000 to 1200 mm.

TAWERS Zi-Tech

Welding technology for zinc-coated steel

Solution to reduce excessive spatter
generation and residual blowholes



Solid wire welding
of zinc-coated steel with less
spatter and blowholes achieved
by our two solutions.

Zi-Active WG4				* TS:Through-arm/External * TM:Separate/Through-arm * TL:External * LA:External
TS	TM	TL	LA	
800	1100	1800	1800	
950	1400	2000		
	1600			
	1800			
	2000			

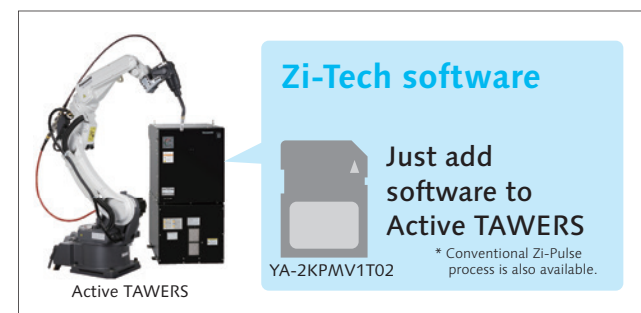
Zi-Pulse WG4/WGH4			
TS	TM	TL	LA
800	1100	1800	1800
950	1400	2000	
	1600		
	1800		
	2000		

Effective for zinc-coated steel welding Reduces spatter and blowholes

Zi-Active

Solution using Active TAWERS

- Uses general welding wire (solid 1.2)
- Applicable range extended to MAG welding in addition to CO₂ welding
- Effective for a wide range of coating weights
CO₂ gas: 45 to 190 g/m²
MAG gas (80:20): 45 to 60 g/m²
MAG gas (90:10): 45 to 60 g/m²



Spatter generation: 75 to 95% reduction
(compared to the conventional CO₂ process)

	Coating weight: 190 g/m ²	
	Normal CO ₂	Zi-Active
Bead appearance	Spatter adhesion: High	Spatter adhesion: Minimal
X-ray	Blowholes: Numerous	Blowholes: Few

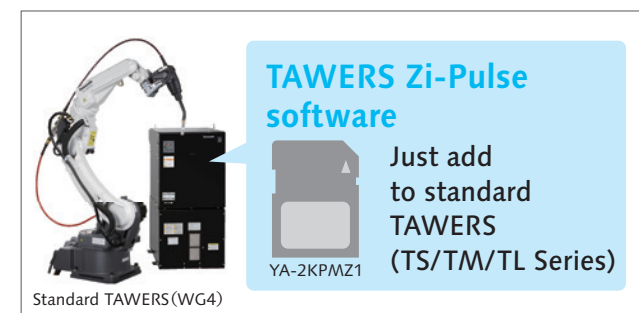
Welding conditions: Wire: YM-50 (φ1.2) Joint: Lap Gas: CO₂
Welding current: 250 A Welding speed: 80 cm/min
Plate thickness: 2.3 × 2.3 mm

Note: Precautions for using AWP4
1. Use a coated pail-pack wire. (Panasonic wire recommended.)
2. Adjust the wire cast diameter to 1000 to 1200 mm.

Zi-Pulse

Solution using standard TAWERS

- Uses general welding wire (solid 1.2)
- Uses MAG gas (90:10) (HD-Pulse welding process)
- Effective for coating weights ranging from 45 to 60 g/m²



Spatter generation: 30 to 60% reduction
(compared to the 80:20 MAG process)

	Coating weight: 190 g/m ²	
	MAG 80 : 20	Zi-Pulse MAG 90 : 10
Bead appearance	Spatter adhesion: Low	Spatter adhesion: High
X-ray	Blowholes: Numerous	Blowholes: Few

Welding conditions: Wire: YM-50MT (φ1.2) Joint: Lap fillet
Welding current: 230 A Welding speed: 80 cm/min
Plate thickness: 2.0 x 2.0 mm

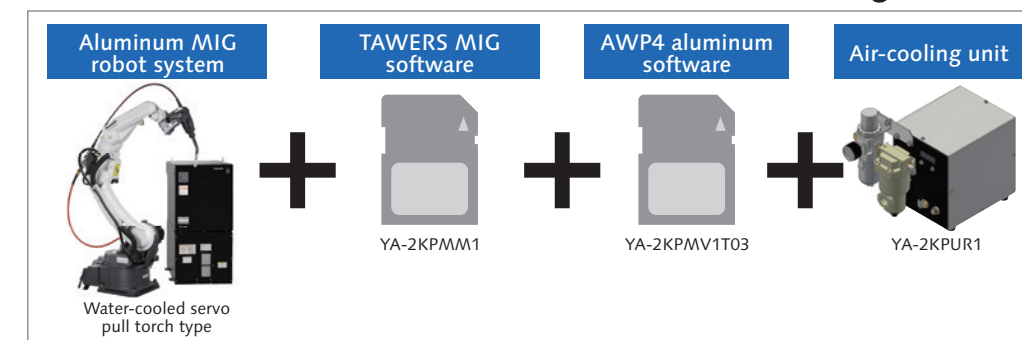
Active TAWERS WG4

Welding technology for zinc-coated steel

Solution to reduce excessive spatter
generation and residual blowholes

S-AWP Aluminum

The ultra-low spatter welding performance of
Active TAWERS is extended to aluminum MIG welding.



Please contact us for details.

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Active TAWERS 4 for aluminum MIG reduces spatter and smut

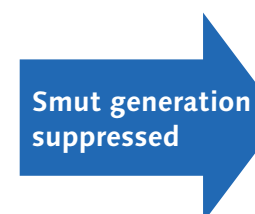
- The ultra-low spatter welding performance of AWP, demonstrated on mild steel, is now extended to aluminum
- A wider current range of 40 to 180 A enables high-speed welding and expansion of applicable plate thickness

Example of medium thickness plate welding (3.0 mm)

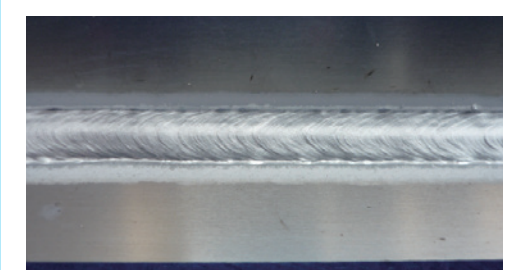
Conventional TAWERS (DC pulse MIG)



Smut formation over the bead



Active TAWERS 4 Aluminum



Shiny bead appearance

Welding conditions: Material: A5052 Joint: T joint Welding current: 155 A Welding speed: 60 cm/min Plate thickness: 3.0 mm

Effective for welding thin aluminum plates

Example of thin plate welding (0.6 mm)



Welding conditions:
Material: A5052 Joint: Butt Welding current: 50 A
Welding speed: 150 cm/min
Plate thickness: 0.6 mm

AC-MIG System

AC control and stable wire feed ensure high-quality
aluminum MIG welding, and powerful output.
Useful for a variety of welding situations.

Additional AC unit increases applications of aluminum MIG welding.

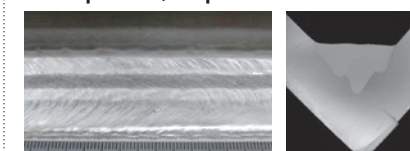
* This system cannot be used in combination with the Active TAWERS aluminum function.

AC unit



Rated output of 350 A Thin to medium and thick plates

One unit can support a wide range of conditions from AC aluminum
welding of delicate thin plates to powerful DC welding of medium and
thick plates. (Output current: 22 A to 350 A)



Joint: Flat fillet
Base metal: A5052
Plate thickness: 15.0 mm
Wire: A5356WY (1.2 mm)
Welding speed: 40 cm/min
Welding current: 280 A DC for 1 pass
250 A DC for 2 to 3 passes

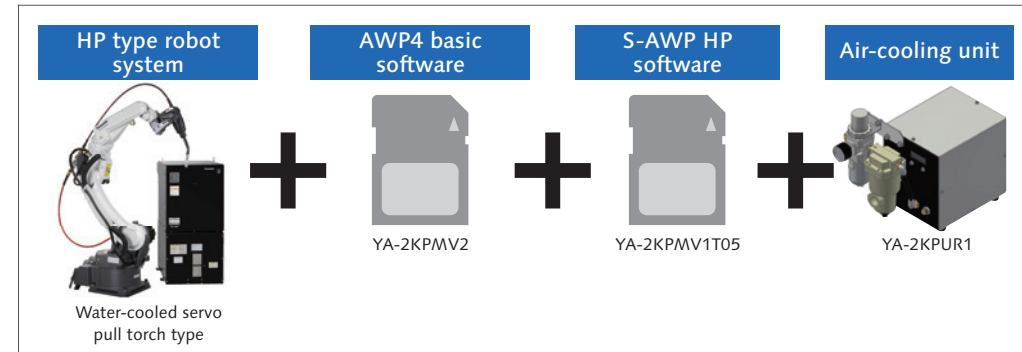
Active TAWERS WGH4

Active Wire Feed Process
available on high-current
range



S – AWP HP

High-speed and medium/thick plate welding achieved with high power



Please contact us for details.

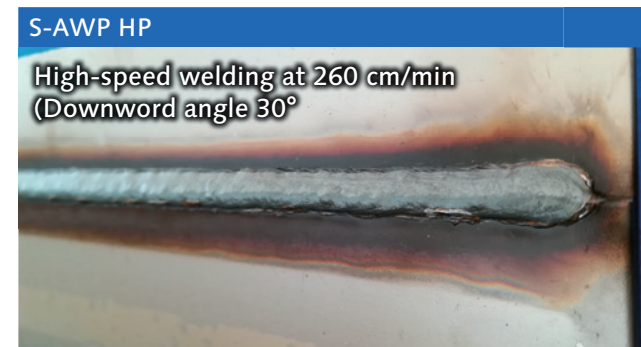
WGH4			
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Even higher-speed welding

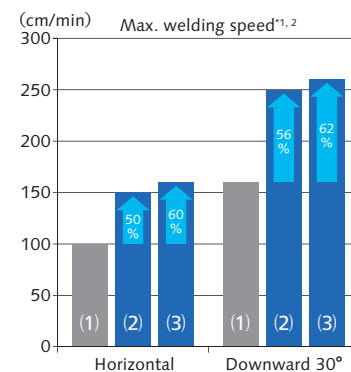
Minimum 50%^{*1} speed increase compared to conventional model



Vertical lap welding
SPCC (1.6 mm), 380 A
YM-50 (φ 1.2), CO₂

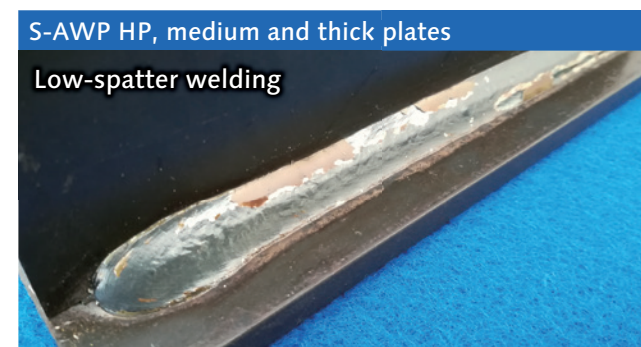
- (1) Active TAWERS 4 standard:
300 A (φ 1.2)
(2) S-AWP HP:380 A (φ 1.2)
(3) S-AWP HP:400 A (φ 1.4)
- *1 Measurements tested in our company's test environment. When you consider the purchase of equipment, verify the suitability for your work at our Process Engineering Center.

*2 Common welding conditions:
Horizontal lap welding, SPCC
(3.2 mm), YM-50 (φ 1.2/φ 1.4), CO₂



Medium and thick plate welding

Minimum 60%^{*1} spatter reduction compared to conventional model



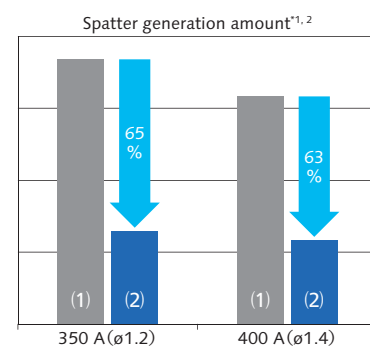
Flat fillet welding
SPHC (9.0 mm)
320 A, 40 cm/min
YM-50 (φ 1.2), CO₂

* SUS-MIG: Applicable only to 350 A or less.
MAG with AWP4: Applicable only to 350 A or less.

- (1) Conventional High Power TAWERS
(2) S-AWP HP

*1 Measurements tested in our company's test environment. When you consider the purchase of equipment, verify the suitability for your work at our Process Engineering Center.

*2 Common welding conditions:
BOP, SPHC (6.0 mm), 100 cm/min,
YM-50 (φ 1.2/φ 1.4), CO₂



Note: Precautions for using AWP

1. Use a coated pail-pack wire. (Panasonic wire recommended.)
2. Adjust the wire cast diameter to 1000 to 1200 mm.

TAWERS[®] WGH4

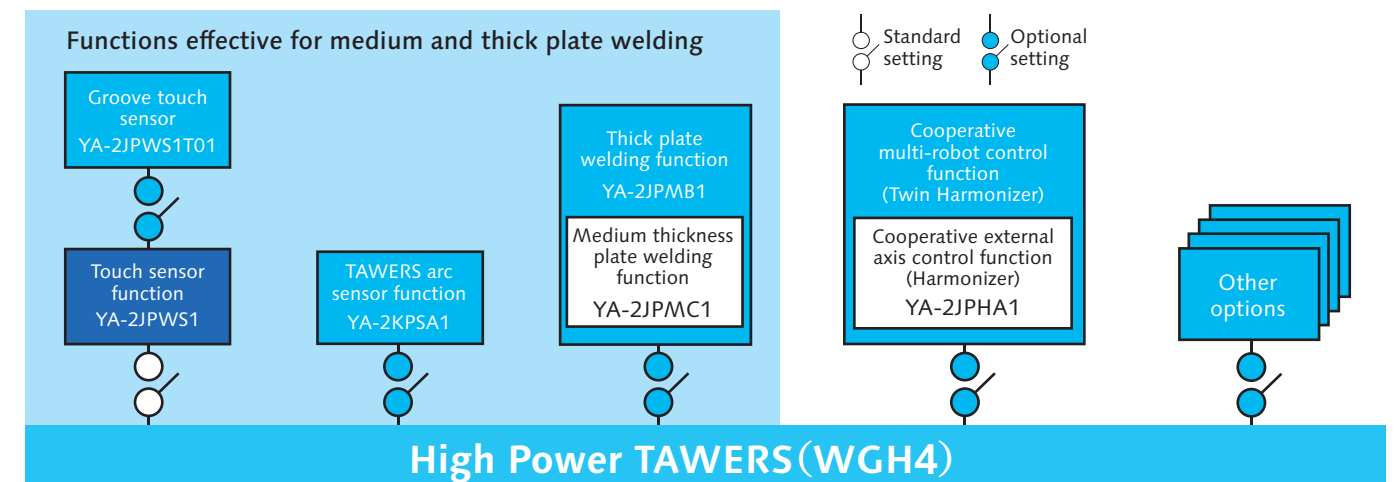
High-power model specialized
for welding medium and thick
plates

TAWERS for medium and thick plates

Various functions can be selected
based on your application

Select necessary options for TAWERS for medium and thick plates.

WGH4			
TS	TM	TL	LA
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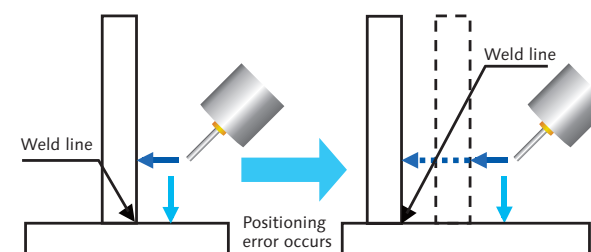
* TAWERS for medium and thick plates: Supplied with touch sensor software and a wire clamp unit

Examples of functions

Touch sensor operation

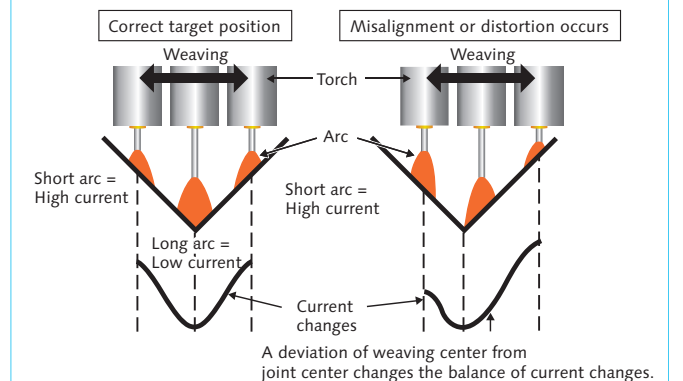
The robot calculates the point of contact with the base metal and determines the weld line.

Measures deviation in the weld line due to errors and determines the weld line again.



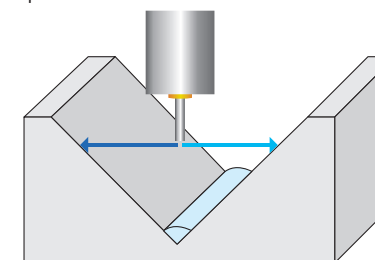
Arc sensor operation

Detects misalignment or distortion of the workpiece and adjusts the position to the correct target position.



Groove touch sensor function

Searches for the groove and detects positioning errors. Senses groove width and center, compensating for misalignment in each workpiece.



Variable weaving function Thick plate welding function (YA-2JPMB1)

Adapts to changes in the groove width. Controls the amount of deposited metal, ensuring uniform bead height.

